NIHR | National Institute for Health and Care Research

NIHR Doctoral Fellowship (DF)

Funding, Support or Development Opportunity	2025/303 NIHR Doctoral Award (Cohort1)

SUMMARY INFORMATION

Application Title	Enabling timely intervention in psoriasis through a patient-held digital stratification tool
Contracting Organisation	King's College London
Partner Organisation(s)	
Start Date	01/06/2026
End Date	31/05/2029
Duration (months)	36
% FTE Commitment	100%
Percentage of award time undertaking practice/clinical work	0%
Total Research Costs Requested	£620,902.58
Has this application been previously submitted to this or any other funding body?	No
Partnership Fellowship	Not Applicable

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NIHR Doctoral Fellowship (DF)

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Dr Manpreet Kaur Sagoo
Organisation	King's College London
Department	St John's Institute of Dermatology
Professional Background	Medically Qualified
Detailed Professional Background	Dermatology
Professional Registration	GMC: 7602295
ORCID	0009-0002-9163-1264
Research Degree Status	Not registered
Date of Registration	
Percentage FTE uptake of research degree	
Date of submission of research degree	
Date of award of research degree	
Contextual Factors	

NIHR Doctoral Fellowship (DF)

APPLICATION DETAILS

Scientific Abstract
Background: Psoriasis is a common, life-long immune-mediated skin disease, affecting 2% of the adult population. Most of the health, societal and economic burden is driven by the 30% of the population who develop severe disease. Highly effective NICE-approved targeted biologic therapies mean that disease remission is now achievable but our 'one size fits all', healthcare centric pathway of care means that access is inequitable, inefficient and delayed. Evidence also suggests that earlier effective intervention may further reduce impact on quality of life and psoriasis-associated multiple long-term conditions (MLTC). Patient-centred strategies to better identify those with severe disease and predict those at risk are now needed to improve access and enable proactive management, maximising the benefits of these high-cost interventions, and thus improve patient outcomes. This proposal builds on our group's existing NIHR-funded smartphone app, mySkin, to address this gap.

Aims and Objectives:
My overarching aim is to develop and validate a patient-held digital stratification tool for psoriasis disease burden.
To do this, I will:
1. Validate patient-driven self-assessment of current psoriasis disease burden, in line with NICE national psoriasis guidelines, using the mySkin platform.
2. Develop and validate a model to predict psoriasis disease burden, integrating clinical, lifestyle and genetic data for future integration into a digital stratification tool.
3. Understand and optimise how current and predicted psoriasis disease burden is communicated to patients using a digital stratification tool, through multi-stakeholder input.

This proposal has been shaped by people with psoriasis, their carers and healthcare professionals who will be involved throughout to ensure inclusive research, relevant to diverse communities.

Methods and Timelines:
Study 1 for Objective 1 (0 – 12 months): This cross-sectional validation study will evaluate patient-clinician concordance between patient self-assessment of current psoriasis disease burden, in line with NICE guidelines digitised on mySkin, compared to standard of care.
Study 2 for Objective 2 (13 – 30 months): This validation study will use large-scale datasets to develop an optimised model to predict psoriasis disease burden (combining clinical, lifestyle and genetic data), using advanced statistical modelling and machine learning approaches. External validation will be performed in diverse datasets.
Study 3 for Objective 3 (11 – 36 months): This qualitative study will use focus groups with people with psoriasis to explore attitudes towards communication of psoriasis disease burden. Focus groups with healthcare professionals will explore attitudes towards patient self-assessment and tool deployment. Insights will inform a communication interface prototype.

Impact:
A patient-held digital stratification tool for self-assessment of current and predicted psoriasis disease burden could enable timely intervention, greater patient autonomy and improved outcomes, through equitable access to care in line with NICE guidelines. This project aligns with NIHR James Lind Alliance research priorities and the NHS Long Term Plan to support digitally-driven, proactive care.

Dissemination:
Findings will be published in open-access journals and presented at national/international conferences, with co-developed Plain English Summaries for international multi-media dissemination via our

established networks.

Plain English Summary

Background: Psoriasis is a common, long-term skin condition that affects 1 in 50 people in the UK. It causes irritating, red, scaly patches on the skin. About 1 in 3 people with psoriasis develop severe psoriasis, which has a serious impact on health, wellbeing and daily life. Although there is no cure, new treatments ("biologics") for severe psoriasis can completely clear the skin. Unfortunately, despite national guidelines, access to these treatments in the NHS is not equal and often delayed.

Research shows that starting biologics at the earliest opportunity can lead to better outcomes, fewer psoriasis-related health problems and improved quality of life. Currently, we cannot predict who will develop severe psoriasis in the future. However, we have recently shown that factors in the environment and in genes can inform who will develop severe psoriasis in the future. At present, there are no reliable tools in the NHS to assess and predict who has or will develop severe psoriasis, to help them get the right treatment at the right time. I will build on our team's existing smartphone app, mySkin (designed with people with psoriasis), to address this gap.

Aim: To develop and test a reliable digital tool that allows people with psoriasis to assess and predict if they have or will go on to develop severe psoriasis.

Methods: First, I will recruit people with psoriasis across the UK from a variety of settings and backgrounds (diverse in age, ethnicity, sex and location) and invite them to self-assess how severe their psoriasis is, on the mySkin app, using health information. I will invite 50 people from this group for an in-person review by a skin specialist to test if they agree with the self-assessment.

Then, using large datasets of diverse populations, I will use different types of mathematical modelling to predict who is likely to develop severe psoriasis. I will test and refine these predictions to improve my model. This will be integrated onto a digital tool in the future.

Finally, I will host group discussions with people with psoriasis (diverse in age, ethnicity, sex and background) to understand how to communicate if a person has or will develop severe psoriasis back to the individual. I will have group discussions with healthcare professionals to understand their views on the tool. I will use this feedback to develop a preliminary platform to communicate results back to individuals.

Working with People and Communities: This project addresses key research priorities highlighted by people with psoriasis. People with psoriasis have directly shaped this project. There was encouraging support for a digital tool to support a personalised treatment approach in psoriasis using health and genetic information. People with psoriasis and the public will advise on this project through 6-monthly meetings. They will help plan and analyse the group discussions. They will also ensure this research is inclusive and relevant to diverse communities.

Impact/Dissemination: This project will support people with psoriasis to get equal access to the right care at the right time, and improve long-term outcomes. It will allow healthcare professionals to provide better care, tailored to individual needs to improve patient experience. I will share my findings in scientific journals and at national/ international conferences. I will work with people with psoriasis to create easily understood materials for circulation via multiple media platforms.

Working with People and Communities Summary

This proposal directly addresses five of the top 20 NIHR James Lind Alliance psoriasis research priorities related to timely intervention and disease progression, to address significant unmet need.

I have co-developed all stages of my project alongside people with psoriasis, their carers and my patient collaborator, Helen McAteer (Chief Executive, Psoriasis Association, the UK's leading psoriasis patient

organisation) with diversity in age, sex, ethnicity, geographical location and psoriasis severity through:
• A 30 minute national Psoriasis Association webinar (n=69 attendees; 193 YouTube views)
• A national survey of people with psoriasis (n=73)
• A 1.5 hour in-person engagement group with people with psoriasis and their carers (n=11)

From these activities, four individuals with psoriasis (diverse in age, sex and ethnicity) have expressed an interest in joining my Patient and Lay Advisory Group (PLAG; total n=6). Additional recruitment will be via the Psoriasis Association and our established PPI network to ensure diversity in views, background and experience. The group will meet biannually, with additional meetings at key decision points (in-person/virtual to maximise inclusivity). To reduce digital exclusion, I will arrange practice online calls and provide phone-in options. All collaborators will be offered formal training via Guy's and St Thomas' NHS Trust and the NIHR Imperial College and Partners Research Support Service Hub. Their input will shape recruitment, data interpretation and dissemination. They will ensure my research is relevant, inclusive and equitable.

All participants in my engagement group and survey strongly supported my research question and felt it was relevant. They emphasised: a) accessing timely intervention for psoriasis is a challenge and b) personalised strategies based on psoriasis disease burden, co-designed with patients, are preferable to a generic approach.

This work has shaped my proposal. Participants in my engagement group were enthusiastic about a digital stratification tool to assess current and predicted psoriasis disease burden. In my survey:
- 78% were willing to provide health information using a digital tool (smartphone app) for personalised psoriasis management advice, shaping the self-assessment approach using health information on mySkin in Objective 1. I will leverage the Psoriasis Association and my PLAG to inform diverse and inclusive recruitment strategies to do this.
- 96% supported the use of genetic data for personalised psoriasis management advice, shaping the inclusion of genetic information in my model in Objective 2. My PLAG will help to interpret model results. External validation in diverse datasets will ensure generalisability and mitigate health inequity.
- 90% of people with psoriasis wanted to know their current and future psoriasis disease burden, shaping the multistakeholder focus groups to understand and optimise this communication through a digital stratification tool in Objective 3. PLAG members will receive qualitative research training, recruit diverse focus groups participants (age, ethnicity, location) and interpret findings.

PPI contributors will receive training to co-develop Plain English summaries for multi-media dissemination, co-author peer-reviewed publications and co-present at national events. PPI contributors will be reimbursed in line with NIHR guidance. GRIPP2 guidelines will be used to report on PPI activities.

SUPPORT AND MENTORSHIP

NIHR Doctoral Fellowship (DF)

Full Name	Professor Sinéad Langan
Organisation	London School of Hygiene and Tropical Medicine
Email Address	sinead.langan@lshtm.ac.uk
Role	Mentor
Description of Support Provided	Professor Sinéad Langan is Professor in Clinical Epidemiology and Medical Director at the London School of Hygiene and Tropical Medicine. She is an honorary Consultant Dermatologist at the St John's Institute of Dermatology at Guy's and St Thomas' NHS Foundation Trust, providing care for patients with chronic inflammatory skin diseases. She is a world-leading researcher in the field of clinical epidemiology and the use of routinely collected health data. As co-Director of Health Data Research UK London (www.hdruk.ac.uk) and co-founder and co-Chair of the REporting of studies Conducted using Observational Routinely collected Data (RECORD) initiative (www.record-statement.org), Professor Langan brings a unique combination of expertise in dermatology, epidemiology, and advanced health data science. Professor Langan has led a large, collaborative, international, programme of work using the power of electronic medical record and administrative data to provide answers for important research questions related to a range of skin diseases. She also co-leads programmes focusing on understanding gene-environment interactions in eczema and is contributing to programmes on stratified medicine in eczema. This makes her an ideal mentor for my NIHR Doctoral Fellowship and as I transition to postdoctoral research. Professor Langan has a track record of directly supervising and mentoring clinical academics at doctoral and post-doctoral level. She will provide a unique research perspective, complementary to the support I will receive from my supervisors. We plan to meet on a regular one-to-one basis, through which she will provide support to pursue my academic and clinical aspirations, and during any challenges I may face. I look forward to learning and working with Professor Langan during this Fellowship.

NIHR Doctoral Fellowship (DF)

Full Name	Professor Catherine Smith
Organisation	GUY'S AND ST THOMAS' NHS FOUNDATION TRUST
Email Address	catherine.smith@kcl.ac.uk
Role	Primary Academic Supervisor
Description of Support Provided	Professor Catherine Smith is a Consultant Dermatologist, NIHR Senior Investigator and Professor of Dermatology and Therapeutics at King's College London. She directs the Skin Therapy Research unit and is a world-renowned psoriasis expert, with experience spanning biomarker discovery to interventional studies. She has strong links to the Psoriasis Association, chaired the NICE psoriasis guideline committee and leads the British Association of Dermatologists biologics guideline committee. She sits on the Steering Committees of the national psoriasis registries (BADBIR, BSTOP). She is involved in major national/international psoriasis collaborations, including the COVID-19 psoriasis registries, and is an International Psoriasis Council board member. She is the Integrated Academic Training Programme Lead at St John's Institute of Dermatology, as well as an NIHR mentor. Professor Smith, Professor Simpson and Dr Mahil have a strong track record of working together in a supervisory team and producing high impact research outputs. Professor Smith will be my primary supervisor for this fellowship. She has been my primary supervisor during my pre-doctoral fellowship and we have established an excellent working relationship. Through her role in the EU BIOMAP consortium, she will support access to key datasets (HUNT, FinnGenn and Estonian Biobank) and networks (the Psoriasis Association, clinical networks) for this fellowship. Her clinical, academic and PPI expertise will be crucial for aligning the digital stratification tool with national guidelines and ensuring clinical relevance. Professor Smith has supervised 7 PhD students to completion, and is currently second/third supervisor for 6 PhD students (including 2 NIHR DRFs). She is currently not supervising any other doctoral students in a primary supervisor capacity. Through her collaborations and networks, she will ensure I have access to relevant career development opportunities during and after this fellowship. She will facilitate the international dissemination of my findings. We will meet weekly throughout the fellowship to discuss progress, troubleshoot emergent challenges/barriers, and ensure my personal training needs are met.

NIHR Doctoral Fellowship (DF)

Full Name	Professor Michael Simpson
Organisation	King's College London
Email Address	michael.simpson@kcl.ac.uk
Role	Academic Supervisor
Description of Support Provided	Professor Simpson is the Head of Department of Medical and Molecular Genetics at King's College London, a renowned centre of excellence in human genetics research. He is an internationally recognised expert in the genetic epidemiology of skin disease, with extensive experience in statistical methods. His work has been instrumental in uncovering the genetic architecture of psoriasis and other dermatological conditions, significantly advancing understanding of the underlying mechanisms of skin diseases. He has substantial experience of developing probabilistic models in academia and also for commercial applications through his former role at Genomics plc where he led an Innovate UK project to develop inference tools for population health management. Professor Simpson will be the second supervisor for this fellowship. He has close collaborations with Professor Smith and Dr Mahil. He has supervised me during my pre-doctoral fellowship and in the development of this application. We have established an excellent working relationship. His direct supervision in this project will ensure rigorous, impactful, and clinically relevant exploration of the multi-variable prediction model combining clinical, lifestyle and genetic information. He will provide formal and informal training in bioinformatics skills, statistics and genetics. He is highly experienced in supervising and mentoring early-career researchers and clinical academics. He has extensive experience of supervising PhD students, including 11 to completion. His PhD students have often been clinicians (5 of total completed) and he has frequently managed students with three or more supervisors. He is currently supervising 3 PhD students, for one he is lead supervisor. Professor Simpson's combined expertise in dermatology genetics, statistical methodologies, and mentorship, positions him as an excellent supervisor for this project. We will meet weekly during lab meetings and fortnightly for one to ones.

NIHR Doctoral Fellowship (DF)

Full Name	Dr Satveer Mahil
Organisation	King's College London
Email Address	satveer.mahil@kcl.ac.uk
Role	Academic Supervisor
Description of Support Provided	Dr Satveer Mahil is a Consultant Dermatologist and Reader at King's Colleg London, with a translational research portfolio focused on using large-scale clinical data and molecular profiling technologies to personalise management strategies to improve outcomes for people with inflammatory skin diseases including psoriasis. Her career to date has followed the NIHR Integrated Academic Training pathway, completing an NIHR Academic Clinical Fellowship which led to an MRC funded PhD and subsequently an NIHR Clinical Lectureship. Post-PhD, she was awarded an MRC-NIHR Clinical Academic Research Partnership Award and she currently holds an NIHR Advanced Fellowship. She is co-director of the COVID-19 psoriasis real-world registries PsoProtect and PsoProtectMe, and is a member of British Association of Dermatologists and European Dermatology Forum psoriasis guideline committees. She is director of St John's DermAcademy, a national dermatology education platform. Dr Mahil is the chief investigator of the mySkin study, a longitudinal cohort study of people with psoriasis exploring determinants of disease onset, progression and treatment outcomes over time, which leverages self-report data collected via the online mySkin platform. She led the co-design of the online mySkin platform and leads an NIHR-funded UK-wide multicentre feasibility trial of personalised biologic treatment strategies for sustaining psoriasis remission, in which mySkin is used to collect longitudinal patient-reported outcome measures and medication use. The mySkin platform will also be leveraged in my proposal. Her expertise in digital health platforms, analysis of real-world data, and mixed methods translational research to enable personalised medicine will be fundamental during all stages of my fellowship. She has supervised 1 PhD student to completion as primary supervisor and is currently primary supervisor to 4 PhD students, including 2 NIHR DRFs. Dr Mahil has supervised me during my pre-doctoral fellowship and I have established an excellent working relationship with her. She will be my fourth supervisor for this fellowship and we will meet every two months to discuss progress.

NIHR Doctoral Fellowship (DF)

Full Name	Professor Sam Norton
Organisation	King's College London
Email Address	sam.norton@kcl.ac.uk
Role	Academic Supervisor
Description of Support Provided	Professor Sam Norton is Professor of Medical Statistics & Applied Health Research at King's College London, with a research focus on the interplay between mental health and long-term physical health conditions, particularly in immune-mediated inflammatory diseases. He has extensive expertise in clinical trials, cohort studies, and prediction modelling, and is a senior statistician with King's Clinical Trials Unit and the NIHR Research Support Service King's & Partners, supporting the development and delivery of a range of NIHR funding applications. Professor Norton has close collaborations with Professor Smith and Dr Mahil, with a strong track record of producing high-impact research outputs. Professor Norton has considerable experience in the development of prediction tools. I have worked closely with Professor Norton to design the data analysis plan for the quantitative aspects of this research proposal and have established a strong relationship. He will be my third supervisor for this project and will directly supervise my training on study design and statistical methods to ensure I gain the necessary skills to perform the analyses required in each phase of the project proposal. He is experienced in guiding early-career researchers and currently supports two doctoral-level NIHR fellows, both of whom are involved with using prediction modelling strategies to inform clinical decision making. Professor Norton has supervised 23 doctoral students, of which 15 have completed and 8 are ongoing. We will aim to meet monthly during the fellowship to discuss progress and ensure my training needs are met.

SUMMARY BUDGET

Justification of Costs

All costs below are at 100% FEC (Cost to Funder). On the budget template, they take into account the 80% Price to Funder contribution.

• Staff Costs

• Salary (£271,522.63)

Full-time salary for lead applicant for duration of the 3-year Doctoral Fellowship, based on nationally agreed NHS pay scales.

• Part time statistician costs 0.1 WTE (£20,999.49)

Funding is sought for a 0.1 WTE statistician to provide statistical oversight and guidance for the predictive modelling component and statistical analysis (which I will be carrying out myself)

• Travel and Subsistence

• Conferences (£3,000)

Conference attendance costs for the British Association of Dermatologists Conference (BAD), the British Society for Investigative Dermatology (BSID) Conference, the European Society for Dermatological Research (ESDR) Conference, and the American Academy of Dermatology (AAD) Conference.

• Placements (£7,500)

Travel/accommodation costs for placements in my Career Development Plan.
• Equipment
• Laptop (£2,500)
As my project involves working with very large datasets (including genetic data) and has a heavy analytical and model-fitting element, I will require a high-performance laptop with a powerful CPU, plentiful RAM and at least 1TB of SSD storage.
• Monitor (£500) and laptop accessories (£100)
Funds for a secondary monitor considering the heavy programming component of my project and appropriate laptop accessories, including a keyboard, mouse and case.
Training and Development
• PhD Student Tuition Fees (£25,310)
• Funds have been sought for the fees (and travel/accommodation where required) for the training courses outlined in my Career Development Plan:
Year 1:
Introductory Course in Epidemiology and Medical Statistics £3,700
Introduction to Qualitative Research Methods £2,000
Public Engagement Course £450
Introduction to Statistics £1,800
Statistical Computing with R £3,390
Clinical Prediction Rules £4,000 (£3,390 + travel/accommodation)
Statistical Methods for Risk Prediction and Prognostic Models £499
Year 2:
Genetic Epidemiology Course £1,100
The King's Fund Emerging Clinical Leaders Course £2,700
Advanced Qualitative Research Methods £4,000
Year 3:
Prognosis Research in Health Care £645
Digital Health Course £1,200
ESDR Future Leaders Academy - £800 inc travel/accommodation
Genomic Medicine Counselling Skills £2000 (£1462 + travel/accommodation)
Knowledge into Action Course £4,000 (£3,390 + travel/accommodation)
Introduction to Implementation Science £1,400
• Research Inclusion (£29,600)
Data access and management for large-scale diverse ancestry datasets for Year 2/3 - Our Future Health (£10,800/year) and East London Genes & Health (£4,000/year)
• PPIE
• Patient and Lay Advisory Group (PLAG) Meetings (£4,800)
Funds for 6 participants to meet for 2 hours (in-person) at 6-monthly intervals with 2 additional meetings at key decision points (£50pp/meeting + £50 pp/meeting for travel reimbursement and refreshments for 8 times during the fellowship, in line with NIHR guidance)
• PLAG co-facilitation of qualitative focus groups (£600)
A trained PLAG member will cofacilitate each focus group for Objective 3 (6 sessions upto 2 hours in-person/online; £50pp/session + £50pp for travel reimbursement and refreshments in line with NIHR guidance)
• Qualitative data analysis training for PPIE collaborators (£160)
Half day online training session two PPIE collaborators (£75 + £5 connection charge, in line with NIHR guidance)
• Qualitative Focus Groups (£3,600)
Payment for qualitative focus groups for Objective 3 (36 individuals for upto 2 hours online/in person at £50pp + £50pp travel reimbursement + refreshments in line with NIHR guidance)
• Payment for in-person clinical validation (£2,500)
Payment for in-person clinical validation for Objective 1 (Total 50 participants for upto 1 hour in person at £25pp + £25pp travel reimbursement, in line with NIHR guidance)
• Other Direct Costs

• Communication interface protoype development (£9,000)
Funds have been requested for the communication interface prototype development in Objective 3
• GSTT NHS Foundation Trust Archiving Fee (£565)
An archiving fee of £565 has been quoted by the GSTT NHS Foundation Trust as part of their role in facilitating participant recruitment for the in-person clinical validation and qualitative work.
• Dictaphone (£475)
For qualitative focus groups
• Focus group transcription costs (£1,206)
Funds to cover transcription costs for focus groups (6 focus groups each upto 2 hours, at £201.00 for 2 hours)
• UK BioBank Dataset Access and Management in Year 2/3 (£7200)
• Data storage, back up and security costs in Year 2/3 (£5000)
• Indirect Costs
• Estates (£96,917.70)
HEI estate costs calculated by the KCL Costing System
• Other indirect costs (£234,639.90)
Indirect cost rate. HEI indirect costs calculated by the KCL Costing System
Note: Our groups has established access to the following datasets with no additional costs: HUNT, Estonian Biobank and FinnGen datasets

NIHR Doctoral Fellowship (DF)

ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	N/A

Lead Applicant:	Manpreet Kaur Sagoo	**Application Number:**	NIHR505922
Contracting Organisation:	King's College London		
Proposed Start Date:	01/06/2026	**End Date:**	31/05/2029

Contributions

Narrative CV template for Personal Career Development Awards
****Lead Applicant / Co-applicant Name:** Dr Manpreet Kaur Sagoo**

****Module 1 – Contributions to the generation of new ideas, tools, methodologies or knowledge****

As an aspiring academic dermatologist, I have a track-record of successfully balancing research pursuits across a project's lifecycle alongside my medical training. This is evidenced by my publication record, which includes five publications (three first-author articles) and three manuscripts in preparation.

Prediction Models: During my King's Health Partners pre-doctoral research fellowship (competitive funding; September 2024 – August 2025), I developed a preliminary psoriasis disease burden prediction model that combines environmental and genetic factors. I grasped key skills in bioinformatic and statistical analyses using R language (including effect-size calculations and enrichment analysis) and generalised linear regression prediction modelling in a large-scale population-based dataset (n=9426; UK Biobank). This preliminary psoriasis disease burden prediction model directly feeds into this doctoral fellowship application.

Genomic Medicine: My ability to interrogate genetic data has been underpinned by my part-time Masters in Genomic Medicine (University of Cambridge; finishing Autumn 2025). Pursuing this alongside clinical dermatology training, this programme has equipped me with a strong understanding of genomic technologies, bioinformatics, and genetic study designs.

Digital Health and Machine Learning: Demonstrating an agile approach to research, my recent contributions have extended to digital health transformation and machine learning. I am co-author on a study describing the potential of a remote monitoring digital health platform to improve understanding of the patient-reported environmental determinants of psoriasis (https://doi.org/10.1093/bjd/ljaf073; February 2025). My role in a recent project harnessing machine learning-based image analysis for the assessment of psoriasis severity has strengthened my skills in building research collaborations (working closely with engineers), curating diverse datasets and image-based modelling. This has culminated in an upcoming first-author national conference presentation (July 2025). Collectively, this builds on my international European Association of Dermatology and Venerology award as the single UK dermatology trainee selected for the Research Summer School (July 2024) where I gained exposure to regenerative skin medicine and artificial intelligence, whilst building international collaborations.

Clinical Trials: As co-investigator for a Phase IIa randomised control trial in adults with allergic rhinitis (August 2022 – February 2023), I acquired expertise in research governance, trial recruitment and research leadership in a multidisciplinary team.

Wet Laboratory: During my intercalated Endocrinology BSc (First Class Honours; 2014-2015), I gained proficiency in cell culture, Western blotting and immunohistochemistry as part of a wet laboratory project investigating diabetic nephropathy. This culminated in a national Kidney Research UK Award, two first-author publications (https://doi.org/10.1007/978-1-4939-9841-8_1,

https://doi.org/10.1016/j.freeradbiomed.2017.12.040), and a national conference presentation (trainee scholarship recipient).

****Module 2 – Contributions to the development of others and maintenance of effective working relationships****

I am committed to fostering effective working relationships and the development of others as an aspiring academic dermatologist and future Principal Investigator.

Supervision and teaching: During the pandemic, by establishing a new collaboration with the British Medical Association (BMA), I pioneered an innovative national medical careers webinar series, benefitting over 1500 medical students (2020 – 2023). As an Intermediate Life Support instructor (2020 – 2024), I have also delivered interactive, simulation-based teaching to multi-disciplinary healthcare professionals and developed strong situational awareness. Lastly, my role as a King's College London medical student examiner (2022 – 2024) has provided me with valuable insight on providing constructive feedback and assessment of progression, while factoring in equality, diversity and inclusion.

Leadership: As the British Association of Dermatologists National Dermatology Trainee Representative (2025) and BMA London Learning Events committee co-chair (2021–2024), I have utilised strategic leadership skills and contributed to the educational development of healthcare professionals. As a local BMA representative during the pandemic, I was awarded a leadership award for my effective advocacy for local resident doctors' concerns and wellbeing. Through these experiences, I have refined my skills in decision-making, negotiation and collaborative problem-solving, which are essential for this doctoral fellowship and my future research career.

Project management: In addition to audits and quality improvement projects, I have spear-headed several dermatology-focussed medical education research projects (including during the pandemic), which have been presented at national and international conferences. This demonstrates my ability to excel in uncertain circumstances and resilience, whilst also motivating other team members. Whilst developing this proposal, I have developed meaningful new collaborations with the Psoriasis Association, contributing to patient webinars and conferences.

Mentorship: In addition to one-to-one mentorship for junior colleagues, I work closely with the King's College London Widening Participation team (2018 – present) to deliver national webinars to promote access to medical school for students from under-represented backgrounds, reaching >2500 students in total.

****Module 3 – Contributions to the wider research community****

Alongside clinical training, I have led 17 national and international research presentations (15 first-author, including 2 international oral presentations) across oncology, stroke medicine, dermatology and medical education. This reflects my broad academic portfolio and wider research community contributions through knowledge mobilisation. As national British Skin Foundation ambassador (2019–2024), I have led medical students in fundraising over £35,000 to directly support vital skin research. Through my work with people and communities, I am committed to championing the NIHR Research Inclusion Framework to improve research equality and diversity. I have

also expressed my interest to peer-review for Clinical and Experimental Dermatology to support development of my critical appraisal skills.

****Module 4 – Contributions to broader society****

I have engaged with diverse sectors of society to facilitate public understanding and knowledge exchange, including presenting my research at Pint of Science (2025) and national patient events for the Psoriasis Association (2024 – present). Other contributions to the broader community include my extra-curricular endeavours during medical school (international and national multi award-winning dancer [2013–2019], King's College London Dermatology Society President [2017-2018)], voluntary medical brigades in rural Central America [2015-2017]). These contributions have been recognised with 5 university awards, culminating in the university's highest honour, the Honorary Lifetime Membership Award (2018). I have also fundraised £5,000 for AIDS Orphan UK by summiting Mount Kilimanjaro.

****Module 5 - Your suitability for this award****

As evidenced by my career to date, I am committed to becoming a future national clinical academic leader in Dermatology. This fellowship is vital to achieve this goal, with an unrivalled opportunity to deliver an ambitious and exciting project. The world-renowned St John's Institute of Dermatology is a centre for excellence in skin research and will provide me with the optimum platform from which to springboard as a clinical academic.

Academic suitability: I have consistently demonstrated the highest standard of academic excellence, demonstrating my ability to excel in this doctoral fellowship and future post-doctoral endeavours. This includes ranking first in dermatology national recruitment, achieving five academic prizes, achieving Distinction in my medical degree (top 10% of cohort; ranked top 3% in pre-clinical years) and First Class Honours in my iBSc.

Research experience: My research career to date has been varied, encompassing wet laboratory experience investigating diabetic nephropathy, through to clinical trial investigator experience and large-scale data bioinformatic analyses. Pursuing a Genomic Medicine Masters at Cambridge University has also equipped me with a robust research skillset, self-discipline and strong organisational skills. This breadth of experience has driven me to develop originality and independent critical thinking as a clinical academic. My engagement in research has garnered me multiple opportunities to present and publish nationally and internationally, whilst cementing my desire to pursue clinical academia. This highlights my commitment to deliver high-impact, meaningful benefit for patients, the public and the healthcare system which I hope to continue through this fellowship.

Step-change in career: I am currently in my second out of four years of dermatology training. Undertaking this doctoral fellowship now is the optimal time to pause my clinical training, affording me an unparalleled opportunity to develop as a clinical academic. Through this fellowship, I will build a diverse research skillset in, but not limited to, public engagement, prediction modelling, qualitative research and digital health transformation. The deliverables from this fellowship will directly underpin my postdoctoral research (for example, through a planned NIHR Academic Clinical Lectureship or NIHR Postdoctoral Award application) as I transition towards becoming a researcher leader in Dermatology. I am committed to delivering real world benefits through my research as a future academic dermatology research leader, supported by my curiosity and drive for life-long learning.

Detailed Research Plan Template for Domestic Programmes and Personal Awards

****Background and rationale****

1. The problem

1.1 Severe psoriasis disease burden is a major health and economic challenge

Chronic plaque psoriasis ('psoriasis') is a common, life-long immune-mediated inflammatory disease (IMID), affecting 2% of the UK adult population[1]. Driven by environmental and genetic factors, it presents heterogeneously with disfiguring, inflamed, red plaques[2]. Most of the health, societal and economic disease burden due to psoriasis is driven by the 30% of individuals who develop severe disease (e.g. those with >10% body surface area involvement)[3, 4]. Severe psoriasis is associated with increasing prevalence of psoriasis-associated multiple long-term conditions (MLTC; psoriatic arthritis, cardiovascular disease, metabolic syndrome), greater cardiovascular mortality, debilitating psychological impact (depression, anxiety) and reduced quality of life, at levels equivalent to common cancers [5-10]. The economic cost exceeds £1 billion/year in the UK[11].

Ultimately, the **more severe the disease, the greater the burden**, with under-represented groups disproportionately affected[9]. This underscores the need for effective stratification by psoriasis disease burden severity to enable timely intervention and improve outcomes for this under-served population.

1.2 Severe psoriasis is under-diagnosed and under-treated, driving inequity

While there is no cure for severe psoriasis, highly effective NICE-approved targeted biologic therapies (available only in specialist care) mean that clear skin and complete disease remission is an achievable goal. Although expensive (£90 million/year for psoriasis in England), these treatments are overall cost-effective[12]. Despite national psoriasis guidelines (NICE CG153; 'NICE guidelines'), access to highly effective biologic treatments for severe psoriasis is highly variable across the UK[12,13]. This **guideline implementation gap** contributes to widening health disparities[12]. Severe psoriasis is under-diagnosed in people with skin of colour, who face disproportionate diagnostic delays and inequitable access to therapeutics[14]. This gap presents a missed opportunity to deliver gold-standard, equitable psoriasis care, tailored to current psoriasis disease burden, which my proposal aims to address.

1.3 The current treatment paradigm in psoriasis care is reactive

Highly effective biologic treatments are only initiated once severe psoriasis is **already established.** This is often too late, once psoriasis-associated MLTC have developed, missing the window to reduce cumulative disease burden. Emergent evidence in psoriasis suggests that timely intervention may modify disease trajectory, offering improved chance of disease remission[15,16]. Currently, progression of psoriasis disease burden is unpredictable. Stratification based on predicted psoriasis disease burden could catalyse a proactive paradigm shift and enable timely intervention.

1.4 There are no validated digital tools for psoriasis disease burden stratification

At present, there are no validated digital stratification tools in routine practice to standardise the assessment of current and predicted psoriasis disease burden, according to NICE guidelines[13]. Such tools have the potential to shift psoriasis care towards personalised digitally-driven care, aligning with the NHS Long Term Plan and the WHO European Digital Health Action Plan[17,18].

To address this gap, in line with the NIHR Health Data Science strategic theme, my project will develop a patient-held digital tool to enable stratification by:

- **Patient self-assessment of current severe disease burden** through digitisation of NICE guidelines to enable timely specialist referral and intervention.
- **Predicted psoriasis disease burden** to enable enhanced monitoring and reduce cumulative disease burden.

2 Importance

2.1 Delivering timely intervention is a key research priority

This project addresses five of the NIHR James Lind Alliance's top 20 psoriasis research priorities, including: the impact of timely intervention (#2), treatment effects on comorbidities (#6), role of genetic and lifestyle factors (#8 and #13) and disease progression (#17)[19]. In my focus group with 15 local primary care professionals, 93% agreed delivering timely intervention for psoriasis is challenging. Top barriers included low patient awareness of psoriasis treatments and difficulty accessing GP appointments. All supported a digital stratification tool to enable timely intervention.

2.2 This proposal has been shaped in partnership with people with psoriasis

My national survey of people with psoriasis (n=73; in collaboration with the Psoriasis Association, the UK's leading patient organisation for people with psoriasis) highlighted significant delays in accessing timely intervention: 44% waited >3 years to see a dermatologist, 60% waited >3 years for effective treatment and 77% reported psoriasis worsening due to care delays. The main barriers were long waiting times for dermatology appointments and difficulty obtaining a referral. These participants have shaped and supported my proposal: **a)** 78% were willing to provide clinical information using a digital tool (smartphone app) for personalised psoriasis management advice, **shaping Objective 1**. **b)** 96% supported the use of genetic information for personalised psoriasis management advice, **shaping Objective 2**. **c)** 90% wanted to be informed of their current and predicted psoriasis disease burden, **shaping Objective 3**. A participant with psoriasis from my engagement group (n=11) stated: *"**more needs to be done** to ensure people with psoriasis have access to the right treatment".* All members of my engagement group supported my proposal of a patient-held psoriasis disease burden digital stratification tool to enable timely intervention.

2.3 Addressing health inequity aligned to NIHR Research Inclusion Framework:

A digital stratification tool for psoriasis disease burden could address existing health inequalities through: **a)** reducing unconscious bias using standardised psoriasis disease burden assessment, aligned to NICE guidelines **b)** democratising access to care across diverse populations, geographical regions and healthcare settings and **c)** enabling efficient resource allocation based on clinical need.

3 Literature Review

3.1 Severe psoriasis disease burden is defined by national psoriasis guidelines

NICE guidelines define severe psoriasis (warranting specialist referral) as inadequate response to topical therapy with significant impact on life (assessed using the Dermatology Life Quality Index) and either: i) ≥10% body surface area involvement or ii) high-impact site involvement (face, scalp, palmoplantar areas and genitalia)[13]. In dermatology clinical practice, the Physician/Patient Global Assessment (clear, almost clear, mild, moderate, severe) and Psoriasis Area and Severity Index score (PASI; ≥10 indicating severe disease) are the gold-standard psoriasis severity measures[13,20.]

3.2 People with psoriasis can reliably self-assess psoriasis severity

In the largest study of patient-clinician concordance of psoriasis severity assessment (n=502), our group demonstrated that individuals with moderate to severe psoriasis can reliably self-assess disease severity in secondary care settings[21]. However, there is a knowledge gap regarding concordance in severity assessment in

community settings. I will address this gap in Objective 1 through clinical validation of patient-led self-assessment of current psoriasis disease burden in the community.

3.3 Timely intervention improves clinical outcomes in psoriasis

Timely intervention with targeted biologic therapy in psoriasis is associated with improved clinical outcomes and disease remission rates[22]. The GUIDE study (n=880) demonstrated faster skin clearance on guselkumab (anti-IL-23p13 biologic agent) in patients with a shorter psoriasis duration (<2 years) compared to long disease duration[15]. Another study revealed improved outcomes in psoriasis patients receiving secukinumab (anti-IL-17A biologic agent) within one year of psoriasis onset[16]. Similar trends have emerged in other IMIDs, such as rheumatoid arthritis and Crohn's disease[23,24]. This underpins the importance of current and predicted psoriasis disease burden stratification (Objective 1 and 2) to enable tailored, timely intervention.

3.4 Genetic liability to psoriasis can predict psoriasis disease burden

Psoriasis has a heritability >60% with 109 known susceptibility loci[25]. The genetic risk factor with the strongest effect is the Major Histocompatibility Complex class I allele HLA C*06:02, influencing psoriasis susceptibility and severity[26,27]. Our group has demonstrated, in the largest psoriasis genetics study, that genetic liability to psoriasis is associated with psoriasis severity at a level equivalent to established individual epidemiological factors[28]. Aligning with the NHS Accelerating Genomic Medicine strategy, this supports the development of a multi-variable prediction tool in Objective 2, utilising genetic information to enable timely intervention before the manifestation of severe disease burden[29]. As most genetics studies are based on **European ancestry datasets**, my project aims to address this distinct gap by prioritising diverse datasets in Objective 2.

3.5 Epidemiological factors also influence psoriasis disease burden

Modifiable and non-modifiable epidemiological factors, including male sex, obesity, smoking and alcohol consumption, are associated with increased psoriasis severity[30-33]. A meta-analysis by our group demonstrated that weight loss interventions can reduce psoriasis severity, reinforcing the role of proactive lifestyle interventions[34]. However, systematic screening for modifiable epidemiological factors is not integrated into routine psoriasis care. The influence of genetic and epidemiological factors **combined** on psoriasis disease burden severity in diverse populations is poorly understood. I will address this knowledge gap through the development and validation of a multi-variable prediction model (Objective 2).

3.6 Stratification tools can inform clinical practice but need validation

Multi-variable stratification of patient cohorts, implemented via stratification tools are well-established in cancer care (e.g. CanRisk) and cardiovascular disease (e.g. QRISK3)[35,36]. Other examples include the ISARIC 4C tool (COVID-19) and the CRASH and IMPACT tools (traumatic brain injury)[37-39]. Although thousands of clinical prediction models are published annually, most lack robust methodology, adequate sample sizes and external validation, rendering them unsuitable for clinical use[40]. No validated models for predicted psoriasis disease burden to support digital stratification tools exist. I will address this gap by further development and validation of my preliminary model (Section 4.2; Objective 2) and multi-stakeholder input (Objective 3).

4. Existing infrastructure, data and preliminary prediction model

4.1 Our NIHR-funded, co-designed mySkin smartphone app ('mySkin')

mySkin is our group's established, co-designed smartphone app for people with psoriasis (available via App Store and Google Play Store). mySkin is currently being implemented in a NIHR-funded UK multicentre trial of personalised dose minimisation strategies in psoriasis remission (NIHR 302258)[41]. The app supports user-friendly data

input for **all variables required in my proposal**. Through our ethically approved UK-wide self-enrolled longitudinal mySkin cohort study, we have recruited an engaged community of 1,000 people with psoriasis (diverse in age, ethnicity and severity)[42,43]. I will leverage this research-engaged cohort during recruitment in my fellowship.

4.2 Preliminary psoriasis disease burden prediction model

I have developed a preliminary psoriasis disease burden prediction model comprising clinical, lifestyle and genetic factors, using UK Biobank (9,426 patients with psoriasis, 1,243 with severe disease) in line with TRIPOD+AI guidelines[44,45]. I applied a validated, dichotomous definition of psoriasis severity using levels of clinical intervention as proxies[46]. Using stepwise feature selection, I derived a minimal, clinically relevant model. Comparative analyses demonstrated this combined approach improves predictive performance over any individual variable group. I have internally validated this model using cross-validation techniques. This model requires further development with **advanced modelling techniques**, plus external validation in **diverse datasets** through Objective 2.

5. Pathway to impact

This timely project aims to deliver a step change in psoriasis care through the development of an intelligent and data-driven psoriasis disease burden digital stratification tool. This could support accelerated referral to specialist care for those with severe current or predicted disease burden to enable timely intervention and equitable access for this population with unmet clinical needs. This project offers the unique opportunity for diverse stakeholders (people with psoriasis, healthcare professionals, charity organisations) to inform future clinical utility and adoption. These insights may translate to other IMIDs and related national health policy.

****Aims and objectives****

Aim: To develop and validate a patient-held digital stratification tool for psoriasis disease burden (Figure 1; page 10). This tool could enable timely intervention tailored to clinical need, greater patient autonomy and improved outcomes.

- **Objective 1: Validate** patient-driven self-assessment of current psoriasis disease burden, in line with NICE guidelines, using the mySkin platform
- **Objective 2: Develop and validate** a model to predict psoriasis disease burden, integrating clinical, lifestyle and genetic data for future integration into a digital stratification tool.
- **Objective 3: Understand and optimise** how current and predicted psoriasis disease burden is communicated to patients using a digital stratification tool, through multi-stakeholder input

****Methodology / Plan****

Study 1 (0 – 12 months)

This will address **Objective 1:** Validate patient-driven self-assessment of current psoriasis disease burden, in line with NICE guidelines, using the mySkin platform.

1.1 Research question: What is the concordance between patient self-assessment of current psoriasis disease burden on mySkin and in-person clinician assessment?

1.2 Study design: This cross-sectional validation study will use validated patient-reported outcome measures from NICE guidelines (including Patient Global Assessment, Dermatology Quality Life Index and Body Surface Area Involvement) for holistic patient self-assessment of current psoriasis disease burden (severe vs non-severe) of individuals in the community, on mySkin[13]. Data input for these measures is already established on mySkin. Leveraging our national networks via the Psoriasis Association and primary care, I will invite people with psoriasis to complete remote self-assessment of current psoriasis disease burden on the mySkin app.

1.3 Patient-clinician concordance of current psoriasis disease burden: I will evaluate patient-clinician concordance between patient self-assessment of current psoriasis disease burden on mySkin, compared to in-person clinician assessment (current standard of care). From the national cohort of people with psoriasis who have completed remote self-assessment on mySkin (Section 1.2), I will recruit and consent a diverse cohort of 50 participants from South-East London for in-person assessment of psoriasis disease burden by a dermatology clinician. The dermatology clinician will be blinded to the patient self-assessment. Using purposive sampling, I will ensure representation across the spectrum of psoriasis disease burden (severe vs. non-severe). I will utilise over-sampling strategies to recruit individuals in whom uptake of internet-enabled health technology remains lower (older age, ethnic minorities, lower socio-economic groups and those with disabilities) to enhance generalisability[47,48]. The concordance in patient-clinician assessment of current psoriasis disease burden will be determined. I will establish factors associated with discordance. Those identified to have severe psoriasis disease burden on clinician assessment will be fast-tracked into our South-East London dermatology care pathway, as per NICE guidelines[13]. Non-severe cases will be provided with educational resources and continued care in the community, in line with NICE guidelines[13].

1.4 Primary outcome: The concordance of patient self-assessed current psoriasis disease burden on mySkin, with evaluation by a dermatology clinician.

1.5 Data analysis: Fleiss' kappa (κ) coefficient will be calculated to compare concordance between the ordered categorical Patient Global Assessment and intraclass correlation coefficient to compare concordance between the continuous Body Surface Area Involvement (%).

1.6 Sample size: The sample size (n=50) is based on the precision of quantitative outcomes. This enables estimates and other proportions with 95% CI +/-15.5%. It provides >80% power to exclude a kappa of 0.4 from the 95% confidence interval when the true value of alpha is at least 0.6–0.7, consistent with substantial agreement.

1.7 Deliverable: Robust clinical validation of patient self-assessment of current psoriasis disease burden as per NICE guidelines, using the mySkin platform.

Study 2 (13 – 30 months)

This will address **Objective 2:** develop and validate a model to predict psoriasis disease burden, integrating clinical, lifestyle and genetic data using large-scale datasets for future integration into a digital stratification tool.

2.1 Research question: How accurately can we predict psoriasis disease burden across diverse populations using clinical, lifestyle and genetic data?

2.2 Study design: This validation study will develop and evaluate the performance of a multi-variable model for predicted psoriasis disease burden using advanced modelling techniques. I will use clinical and lifestyle measures close to the time of first psoriasis diagnosis (from electronic health record data), alongside a polygenic score for psoriasis susceptibility that our group have previously demonstrated is also associated with psoriasis severity. Input variables will include sex, age of psoriasis onset, smoking status, alcohol consumption, adiposity measures and genetic information. I will use our existing approach using levels of clinical intervention as a proxy for psoriasis severity (Background Section 4.2)[46]. This will build on my preliminary psoriasis disease burden model (Background Section 4.2), though will consider the temporal relationship between clinical and lifestyle measures and the onset of severe disease. External validation in diverse populations will be performed. This study will be conducted and reported as per TRIPOD+AI guidelines, with

prediction modelling expertise from my supervisors (Professor Norton, Professor Simpson) and collaborators (Dr Modat)[45].

2.3 <u>Datasets:</u> I will use deeply phenotyped and genotyped large-scale diverse datasets for model development and validation, which have availability of all variables required for this study[44,49-54].

Dataset	Country	Ancestry	Psoriasis Cases (n=)		Harmonised variables?	Access Status	Planned use
			Total	Severe (%)			
UK Biobank	UK	Predominantly White British	9,426	1,243 (13.2%)	Yes	Confirmed	Development & internal validation
East London Genes & Health	UK	100% British Bangladeshi & Pakistani	1,830	Not yet classified	Yes	Provisional confirmed	External validation & generalisability
Our Future Health	UK	White 90.1% Asian 5.3% Black 1.5% Mixed 1.7% Other 1.3%	10,393	Not yet classified	Yes	Application in progress	
Estonian Biobank	Estonia	100% White European	14,167	1,639 (11.6%)	Yes (via BIOMAP Consortium, led by Professor Smith)	Confirmed	External validation & transportability
FinnGen	Finland		12,707	5,365 (42.2%)			
Trøndelag Health Study (HUNT):	Norway		8,603	1,491 (17.3%)			

2.4 <u>Inclusion criteria:</u> Adults (≥18 years old) with psoriasis and longitudinal psoriasis disease burden data for classification of severe vs non-severe (using levels of clinical intervention as proxies for severity).

2.5 <u>Exclusion criteria:</u> Individuals with insufficient/missing healthcare data not appropriate for multiple imputation e.g greater than 50% missing data or longitudinal data not available or classification of disease burden not possible.

2.6 <u>Data analysis</u>: Model development will follow a two-stage process using an 80% training / 20% internal validation split within UK Biobank, building on my preliminary model (Background Section 4.2)[44]. Initial modelling will build on my previous analyses and include traditional approaches such as generalised linear models and penalised regression techniques (e.g. Lasso, Ridge) to address multicollinearity and overfitting. More complex machine learning models will then be applied and compared, including Random Forests, Support Vector Machines, and XGBoost. Model performance will be assessed using measures of discrimination (C-statistic) and calibration (calibration slope, calibration-in-the-large). Clinical utility will be evaluated using receiver operating characteristic curves to assess sensitivity and specificity and precision-recall to assess positive predictive value and sensitivity. In parallel, a simplified clinical model will be developed using only readily available clinical and lifestyle data, to enhance feasibility of implementation in real-world clinical settings. Missing baseline data will be accounted for by using multiple imputation by chained equations, and models will be fitted across imputed datasets with performance metrics pooled to ensure robustness. Complete case analysis will be conducted as a sensitivity analysis.

Additional, more exploratory analysis, will attempt to account for disease progression over time using longitudinal modelling techniques. I will first apply a linear mixed model as a baseline approach. I will then use Gaussian process regression to model individual disease trajectories with measures of uncertainty, key for clinical tool development. Deep learning approaches, particularly Long Short-Term Memory (LSTM) neural networks, will also be explored for their ability to capture temporal patterns from longitudinal data with automatic feature learning.

2.7 Model validation: External validation will be performed using large-scale datasets of diverse cohorts (Section 2.3): the East London Genes and Health cohort (enriched for ethnic minority populations), the national Our Future Health study, and European datasets from the European (EU) BIOMAP consortium[49-54]. Harmonised variables, overlap with UK Biobank data fields and large sample sizes make these datasets ideal to develop an accurate and calibrated prediction model across diverse ancestries.

2.8 Sample size: Using 80% of the UK Biobank cohort for model training (c. 7500 people with psoriasis; 990 with severe psoriasis), the Riley et al. approach (pmsampsize package), indicates the sample size provides sufficient precision for model development, allowing for over 90 candidate predictors while maintaining model shrinkage and calibration[55]. This confirms sufficient power for robust modelling.

2.9 Deliverable: An externally-validated multi-variable model (combining clinical, lifestyle and genetic information) for predicted psoriasis disease burden ready for future integration into a digital stratification tool (available to the scientific community).

Study 3 (11 – 36 months):

This will address **Objective 3:** Understand and optimise how current and predicted psoriasis disease burden is communicated to patients using a digital stratification tool, through multi-stakeholder input

3.1 Research question: How can current and predicted psoriasis disease burden be communicated to patients using a digital stratification tool?

3.2 Study design: This qualitative study will explore acceptability of current and predicted psoriasis disease burden communication to patients using a digital stratification tool, through multi-stakeholder focus groups[56,57]. Focus group participants will include people with psoriasis (recruited from Study 1, the Psoriasis Association and national primary/secondary care networks) and healthcare professionals (dermatologists/primary care professionals; recruited via our well-established research networks). I will utilise over-sampling strategies using our group's co-designed "skin research inclusion toolkit" to recruit individuals in whom uptake of internet-enabled health technology remains lower (older age, ethnic minorities, lower socio-economic groups and those with disabilities) to enhance digital inclusion[47,48]. All participants will be reimbursed[58].

3.3 Focus groups: A total of six focus groups (two hours each; six participants/group; online/in-person) will be conducted and co-facilitated by a trained member of my Patient and Lay Advisory Group. A Patient and Public Involvement (PPI)-informed topic guide and general questioning will be used to achieve convergence to group norms[56]. Separate sessions will be held for each stakeholder group for open discussion among those with shared experiences (healthcare professionals and people with psoriasis; 3 focus groups for each group)[56]. Discussion will focus on:

For people with psoriasis: a) Attitudes to communication of current and predicted psoriasis disease burden **b)** Scoping of acceptable thresholds for clinical intervention for predicted psoriasis disease burden through scenarios, to inform further post-doctoral research **c)** Research priorities for postdoctoral research and clinical practice

For healthcare professionals, this will include all the above and: a) Attitudes regarding patient self-assessment **b)** Acceptable levels of model performance for clinical use **c)** Scoping of perspectives on integration into existing NHS care pathways

Discussions will be audio-recorded, transcribed verbatim to enable inductive thematic analysis using NVivo software. A systematic approach will be used to organise collected data into relevant codes, which will be grouped into potential themes using an iterative process[59]. I will involve two PPI collaborators in the data

analysis to minimise bias. This will guide the iterative refinement of inclusive, acceptable communication of current and predicted psoriasis disease burden.

3.4 Prototype for current and predicted disease burden communication: I will explore formats for communicating current and predicted psoriasis disease burden (e.g. written summaries, traffic light visuals, graphs) tailored to diverse end-user needs. Outputs may include burden summary (severe vs non-severe) and guideline based recommendations (specialist referral for severe cases vs education and community care for non-severe cases)[13]. Using iterative feedback, I will develop a prototype for psoriasis disease burden communication for a digital stratification tool, in collaboration with The Hyve (www.thehyve.nl), our specialist software developers, aligned with the Department of Health and Social Care digital standards[60].

3.5 MHRA regulatory readiness planning: I will work with KCL's London Institute for Healthcare Engineering for the expertise in future MHRA medical device readiness[61].

3.6 Deliverable: A prototype of a psoriasis disease burden communication feature for a digital stratification tool, informed by multi-stakeholder input.

3.7 By the end of my fellowship: I will have developed and validated a digital stratification tool for psoriasis disease burden. In my post-doctoral research (e.g via a NIHR Academic Clinical Lectureship, NIHR Postdoctoral Award), I will explore a regional pilot to test the utility of patient-driven access to care through this digital stratification tool and progression to MHRA medical device approval.

4. Working in partnership with people and communities: This proposal has been shaped by people with psoriasis (Background Section 2.2). During my project, I will leverage PPI through Helen McAteer (my patient collaborator**;** Chief Executive of the Psoriasis Association) and my Patient and Lay Advisory Group (Working with People and Communities summary) who will assist with diverse participant recruitment, interpretation of results, multi-media dissemination of findings and co-development of lay summaries. I will report PPI activity in publications (GRIPP2 guidelines) and acknowledge all contributors during dissemination[62]. I will perform quality evaluation of all PPI activities using a PPI impact log, anonymous feedback and GRIPP2 short form[61]. Specific involvement includes: **Study 1:** Recruitment of a diverse cohort for remote/in-person self-assessment. **Study 2:** Review of prediction model, interpretation of prediction model results**. Study 3:** Co-facilitation of focus groups with flexible free training offered, co-development of topic guides for focus groups, involvement in qualitative data analysis with flexible bespoke training offered.

5. Research Inclusion: My proposal prioritises tackling existing health inequalities, aligned to the NIHR Research Inclusion Strategy. **Study 1 and 3:** Over-sampling strategies will be employed, using our group's co-designed "skin research inclusion toolkit" to recruit diverse individuals in whom uptake of internet-enabled health technology remains lower (older age, ethnic minorities, lower socio-economic groups and those with disabilities) to enhance digital inclusion and. Reimbursement will be offered as per NIHR guidance including childcare/carer/internet costs to maximise research inclusion. **Study 2:** Prioritisation of model external validation in diverse datasets: Our Future Health (representative of the UK population) and East London Genes & Health (100% South Asian ethnicity) to ensure inclusivity.

****Knowledge mobilisation, dissemination and impact****

1. Impact: a) Patient: A patient-held digital stratification tool for current and predicted psoriasis disease burden can empower individuals by enabling timely intervention, greater autonomy and improved outcomes. **b) Clinician:** Enables holistic, efficient assessment to facilitate timely referral. Offers proactive and personalised monitoring and intervention for those at high-risk of severe disease. Enables the opportunity for

management of modifiable risk factors (e.g. obesity, smoking, alcohol). **c) Research:** Cross-disease and cross-disciplinary research for assessment and prediction of multifactorial disease burden could be informed. **d) Commissioner:** Supports more equitable, needs-based service design and efficient NHS resource use.
2. Dissemination: a) Conferences: I will present my findings at national/international clinician/scientist/patient conferences: British Society for Investigative Dermatology Conference, British Association of Dermatologists Conference, Psoriasis Association Conference, European Society for Dermatological Research Conference and American Academy of Dermatology Meeting. **b) Publications**: Three peer-reviewed open-access publications: **i)** Current psoriasis disease burden self-assessment using a digital tool **ii)** Predicted psoriasis disease burden model **iii)** Multistakeholder attitudes towards a digital stratification tool for current and predicted psoriasis disease burden. I will work with PPI collaborators to produce multi-media Plain English Summaries. **c) Health Policy**: My results will inform UK guidelines updates via my publications and supervisors. Professor Smith chaired the NICE psoriasis guideline committee. Digital engagement insights across ancestries will be shared with global networks (International Psoriasis Council), which could inform equality guidelines.
****Project management****
1. Fellowship oversight: I will lead this project, supported by my supervisors who will provide academic, clinical and methodological oversight. We will meet regularly to ensure my research and training progresses according to the Gantt chart. My progress will also be monitored at King's College London via 6-monthly meetings with an independent Progression Committee. **2. Patient and Lay Advisory Group:** I will meet this group (n=6) 6-monthly (additional meetings at key points) to discuss PPI strategy, gain feedback on progress, and plan results dissemination and implementation. **3. Ethics:** Ethical approval is in place for patient digital self-assessment in Study 1 via our mySkin study (REC22/NI/0193). I will submit an ethics amendment for the in-person assessment and obtain ethical approval for focus groups in Study 3. **4. Data protection and confidentiality:** All data will be stored in compliance with the Data Protection Act 1998 and GDPR. Access will be restricted to the research team. **5. Success Criteria: a)** Delivery of research to time and target. **b)** Peer-reviewed publications (Dissemination Section b) **c)** Presentations to clinical, academic, lay and industry audiences. **d)** Completion of Career Development Plan. **e)** Development of competitive postdoctoral award application. **6. Risks: Study 1:** Failure to recruit sufficient diverse participants. I will work closely with Helen McAteer (Psoriasis Association) and my PLAG to support inclusive recruitment from diverse under-represented communities. **Study 2:** Datasets of European ancestry may limit model generalisability. To mitigate this, I have prioritised external validation in diverse UK datasets (Our Future Health, East London Genes & Health). **Study 3:** Technical and regulatory challenges to meet MHRA standards. I will collaborate with KCL's London Institute for Healthcare Engineering and software engineers to mitigate this. **7. Collaborators: a)** Helen McAteer (Chief Executive, Psoriasis Association) is my patient representative lead and has reviewed the Plain English Summary. She will oversee recruitment strategies, chair PLAG meetings and review dissemination materials. **b)** Dr Marc Modat (Reader, KCL School of Biomedical Engineering & Imaging Sciences) will provide machine learning expertise for Objective 2.
****Intellectual property and commercialisation**** I will work closely with Commercial Innovation, the Technology Transfer Office at GSTT and KCL, who will support with reviewing, assessing, protecting and commercialising any relevant new intellectual property arising from the study.

Figure 1: NIHR Doctoral Fellowship

Existing infrastructure & data	myskin Our group's established NIHR-funded mobile app (NIHR 302258)	Preliminary psoriasis disease burden model (developed in pre-doctoral fellowship)	National psoriasis networks & medical device development expertise
Objective	**1. Validate** patient-driven self-assessment of current psoriasis disease burden, in line with NICE guidelines, using the mySkin platform	**2. Develop and validate** a model to predict psoriasis disease burden, integrating clinical, lifestyle and genetic data for future integration into a digital stratification tool	**3. Understand and optimise** how psoriasis disease burden is communicated to patients using a digital stratification tool, through multi-stakeholder input
Study	**Study 1: Cross-sectional validation** study of patient-clinician concordance of current psoriasis disease burden assessment (n=50)	**Study 2: Validation study** to develop and evaluate a multi-variable prediction model using large-scale datasets	**Study 3: Qualitative study** using focus groups with people with psoriasis and healthcare professionals (n=36)
Deliverable	**Robust clinical validation of patient self-assessment** of current psoriasis disease burden as per NICE guidelines, using the mySkin platform	**Externally validated** model for predicted psoriasis disease burden for future integration into a digital stratification tool	**Prototype** of a psoriasis disease burden communication feature for a digital stratification tool
Research Inclusion	**Over-sampling strategies** to recruit diverse individuals (older age, ethnic minorities, lower socio-economic groups and those with disabilities) for digital inclusion	Prioritisation of model external validation in **diverse datasets** to ensure inclusivity	**Over-sampling strategies** to recruit diverse individuals (older age, ethnic minorities, lower socio-economic groups and those with disabilities) for digital inclusion
Working with people & communities	Patient and Lay Advisory Group to support **recruitment** of a diverse cohort	Patient and Lay Advisory Group to help **review and interpret** prediction model results.	Patient and Lay Advisory Group to **co-facilitate** of focus groups, **co-develop** topic guides and help with qualitative data **analysis**. Flexible training offered

Postdoctoral Research	**Regional pilot implementation of digital stratification tool for psoriasis disease burden**	**MHRA regulatory ready digital stratification tool medical device**

Career Development Programme for Personal Career Development Awards

I have designed a bespoke training and career development programme, closely with my supervisors to ensure I have a multi-disciplinary skillset to successfully deliver the technical, clinical and translational components of my proposal and support my development as an academic dermatologist.

****Training courses****

Statistics and Computer Programming: I have completed an introduction to bioinformatics module as part of my Masters in Genomic Medicine, providing me with a grounding in R programme and RStudio. As Objective 2 of my project relies on large-scale population datasets and advanced modelling techniques, I would benefit from further formal training to build my proficiency. I will consolidate these skills to develop advanced competencies starting from the first year of my fellowship through the five-day **Introduction to Statistics with R Course** (University College London) and the eight-week online part-time **Statistical Computing for Health Care Research Course** (University of Oxford). These courses will strengthen my ability to apply statistical principles, advanced statistical and machine learning modelling methods to refine my prediction model in Objective 2. To support population-level data interpretation, I will attend the two-week online **Introductory Course in Epidemiology and Medical Statistics** (London School of Hygiene and Tropical Medicine). This will enhance my understanding of confounding, effect modification, and interpretation of statistical outputs in the context of multi-variable health data for my prediction model.

Prediction Modelling: This project centres on developing a clinically useful prediction model for future psoriasis disease burden in Objective 2. To ensure methodological rigour in advanced modelling techniques, external validation studies and clinical utility, I will attend the three-day online **Statistical Methods for Risk Prediction and Prognostic Models course** (University of Birmingham) and the eight-week part-time online **Clinical Prediction Rules** course (University of Oxford) at the end of my first year. This will ensure my prediction model study is conducted in line with best practice and TRIPOD+AI guidelines. In my third year, this will culminate in attending the three-day **Prognosis Research in Healthcare: International Summer School** (University of Keele) to incorporate the latest advances into model refinement. This work will be supported by my supervisor Professor Sam Norton and collaborator, Dr Marc Modat.

Genetics: As my model in Objective 2 integrates genetic data, I will undertake the five-day online **Genetic Epidemiology Course** (University of Bristol) in my second year to develop specialist skills in polygenic score construction and interpretation. I will complete the five-day **Genomic Medicine Counselling Skills course** (University of Cambridge) in my third year to enable me to understand how to effectively communicate genetic risk in clinical and patient-facing settings to help inform strategies in Objective 3. This work will be supported by dedicated supervision from Professor Michael Simpson, an expert in complex disease genomics.

Qualitative training: To enable effective qualitative data collection and analysis during Objective 3, I will complete the five-day **Introduction to Qualitative Research Methods** (University of Oxford) course in my first year and subsequently build on this through the five-day **Advanced Qualitative Research Methods** (University of Oxford)

course in my second year. I will build on this by completing a part-time **NVivo Core Skills Online Course** (NVivo Academy) in my second year also. These will equip me with knowledge on how to successfully conduct focus groups, and give me practice with coding and thematic analysis. This will enable me to design and lead high-quality qualitative research to understand attitudes towards a digital stratification tool among patients and clinicians in Objective 3.

Patient and Public Involvement: My project is grounded in PPI, from defining the problem to shaping the development of a prototype communication interface in Objective 3. During the first year of my fellowship, I will attend the **Online PPI training** for researchers (NIHR Cambridge Biomedical Research Centre PPI team). I am keen to attend the sessions on ethnicity inclusion training, building and maintaining a PPI group, using PPI to help communicate research and evaluating PPI. In addition, I will build on my skills in public communication by attending a one-day **Public Engagement course** (Science Museum) in my second year. I will work with the Psoriasis Association throughout the project to support ongoing patient involvement and engagement.

Digital Health Transformation: In my third year, I will complete a four-week online **Digital Health course** (University of Harvard; 3 hours per week). This will provide insight and knowledge for digital tool development, evaluation, and deployment, supporting Objective 3 and my postdoctoral work.

Implementation Science: At the end of my third year, I will complete the five-day **Knowledge into Action course** (University of Oxford) and the online **Introduction to Implementation Science Modules** (NIHR). These will build my skills and knowledge in intervention adaptation, stakeholder engagement and evaluating real-world impact in diverse populations, core to embedding my tool in NHS care pathways, building on Objective 1 and Objective 3. This will be a useful foundation for my postdoctoral research.

Leadership: To prepare for future research leadership roles, I will complete the six-day online **Kings Fund Emerging Clinical Leaders Course** in my second year. This will support the development of strategic leadership, team management skills and interdisciplinary collaboration. I will attend the three-day **Future Leaders Academy** (European Society for Dermatological Research) in my third year, which focuses on delegation, conflict resolution, and influencing strategic change. This presents a unique opportunity to connect with future leaders in dermatology and receive mentorship, greatly benefitting my postdoctoral career.

****Placements****

External UK Placement: I will undertake a two-week placement with the Dermatology Research Department at the University of Manchester at the end of Year 1 to explore future opportunities for further in-person validation of patient self-assessment of psoriasis disease burden in geographically and ethnically diverse UK populations.

International Placement: I will undertake a one-month placement with Professor Justin Ko, Director and Chief of Medical Dermatology at Stanford University (USA). He currently spearheads the Stanford Dermatology Departments digital health strategy. He has an interest in improving care delivery through leveraging machine learning and artificial intelligence. This will be a valuable opportunity to learn from his expertise in dermatology digital health to inform my digital stratification tool development, optimisation and implementation.

Building Collaborations

I will maintain and expand international collaborations with the HUNT Study, Estonian Biobank and FinnGen study groups, which are central to the external validation phase of my project in Objective 2. I will also develop new links with Our Future Health and East London Genes & Health to explore broader application of my prediction model in UK populations with diverse ancestries through Objective 2. I will develop strengthened collaborations with Stanford University through my international placement to support the further development of my digital stratification tool in Objective 3 and future post-doctoral work.

Conferences

To disseminate findings and build professional networks, I will present at national and international conferences including:

- British Society of Investigative Dermatology (Year 1)
- Psoriasis Association Conference (Year 1)
- European Society for Dermatological Research (Year 2)
- British Association of Dermatologists Annual Meeting (Year 2)
- American Academy of Dermatology (Year 3)

Other

I will be hosted by the St John's Institute of Dermatology, a world-renowned centre of excellence in skin research with an outstanding track record of supporting clinical academic PhD fellows. The KCL Doctoral Development Programme offers a comprehensive suite of training opportunities, including courses on effective communication, teaching methods, and career management. I will engage in mentoring schemes through the NIHR Academy, supporting my transition to post-doctoral research and leadership.

School of Basic & Medical Biosciences

Prof John McGrath

Head of Department

St John's Institute of Dermatology

St John's Institute of Dermatology

9th Floor, Tower Wing

Guy's Hospital

London SE1 9RT

sjid.oo@kcl.ac.uk

9th June 2025

NIHR Doctoral Fellowship Application - Dr Manpreet Kaur Sagoo
Head of Department Letter of Support

My name is John McGrath, and I am the Academic Head of St John's Institute of Dermatology within King's College London (Guy's Hospital campus). It is a pleasure to be asked to write a supporting statement for Dr Manpreet Kaur Sagoo's NIHR Doctoral Fellowship Application.

I confirm that St John's Institute of Dermatology, part of King's College London (KCL), supports this application and that, if funded, the research and training will be supported and administered by KCL. I confirm that Manpreet Kaur Sagoo will undertake this work and is eligible, for the duration of the award, to live and work in the UK.

Dr Sagoo wishes to develop a clinical academic career and I confirm that I am aware of the principles and obligations published by NIHR to support her clinical academic training and I confirm that KCL is committed to these principles and obligations.

The St John's Institute of Dermatology (School of Basic and Medical Biosciences, KCL) is an internationally renowned centre for the delivery of innovative clinical care for patients with skin diseases, cutting-edge education, training, and research. A recent independent RAND assessment ranked St John's as the leading clinical and research institution for dermatology in the UK. St John's occupies 1,200 square metres of state-of-the-art laboratory space (Guy's Hospital campus) and is fully integrated into the Faculty of Life Sciences & Medicine's School of Basic & Medical Biosciences. An active programme of graduate education is pursued including research training (PhD and MD) and a world-renowned taught course (MSc) in Clinical Dermatology. The Institute excels in research training, with a >95% success record of its Integrated Academic Training Pathway Fellows obtaining external funding from MRC/NIHR. In addition, St John's Institute of Dermatology plays a leading role in the training of dermatologists, both in the UK and around the world. Research at St John's targets 4 key areas: inflammatory and autoimmune skin disorders (the key area of this research proposal), cutaneous oncology, genetic skin disorders, and skin-environment interactions. Our research is supported by grants from a range of sources including NIHR, MRC, The Wellcome Trust, EU, US Foundations, British Skin Foundation, national charities, and commercial sources. Our department research income averages >£3.5m per year (total value of current grants £17m) with a School average research grant income of >£25m per year. The latest submission to the REF confirmed Clinical Medicine was independently rated to have 90% of its research as world leading or internationally excellent (3 or 4*) and 84 % of impact as world leading (4*). King's College London is part of King's Health Partners, one of the UK's five Academic Health Sciences Centres, where world-class research, teaching and clinical practice are brought together for the benefit of patients. The three NHS partners; Guy's and St Thomas', King's

College Hospital (KCH) and South London and Maudsley NHS Foundation Trusts, are among the highest rated Trusts in the country and compete on the world stage and attract the best clinical academics to pioneer world-leading advances in health and well-being.

As a host organization, KCL is committed to creating and maintaining an inclusive and supportive research culture for all, championing equality, diversity and inclusion, and research integrity. KCL has been a member of the Athena SWAN Charter since 2007 and in 2021 it received a Silver Award. In 2018, the Faculty of Life Sciences & Medicine was recognised with a Faculty Athena SWAN Silver award (renewed in 2024 and valid to 2029) as recognition of our ongoing commitment to achieving gender equality and an inclusive environment.

Dr Manpreet Kaur Sagoo will have full support from the extensive infrastructure within the Department, School, KCL, and King's Health Partners. We will ensure the equipment and facilities successfully allow Manpreet to pursue her research. We will also provide extensive personal support and mentorship to allow Manpreet to fully develop her research ideas, and also to embed her career as a future leading clinical academic in dermatology.

Dr Sagoo's project titled "Enabling timely intervention in psoriasis through a patient-held digital stratification tool" aims to develop and validate a patient-held digital stratification tool for psoriasis disease burden. This has the potential to facilitate more personalised, proactive management for individuals with psoriasis and improve patient outcomes. The project has three main workstreams:

1. Validate patient-driven self-assessment of current psoriasis disease burden, in line with NICE national psoriasis guidelines, using the mySkin platform
2. Develop and validate a model to predict psoriasis disease burden, integrating clinical, lifestyle and genetic data for future integration into a digital stratification tool
3. Understand and optimise how current and predicted psoriasis disease burden is communicated to patients using a digital stratification tool, through multi-stakeholder input

This work is integrated with the clinical academic activities of two Faculty members within the department, Professor Catherine Smith and Dr Satveer Mahil, who will provide supervision, training and mentoring.

In short, I am delighted to offer this Head of Department letter of support and full institutional support for this application.

Please do not hesitate to contact me if any further information is required.

Yours sincerely,

Professor John McGrath MD FRCP FMedSci

Professor of Molecular Dermatology
Mary Dunhill Chair in Cutaneous Medicine
St John's Institute of Dermatology
Head of Department, St John's Institute of Dermatology
john.mcgrath@kcl.ac.uk

NIHR Doctoral Fellowship Timeline:		Months											
		0-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24	25-27	28-30	31-33	34-36
Project Work													
Objective 1	Develop protocol and submit ethics amendment												
	Participant recruitment												
	In-person clinical validation												
	Write up and submit publication												
Objective 2	Obtain access to datasets												
	Data transfer and preparation												
	Model development in UK Biobank												
	Model validation (internal and external)												
	Write up and submit publication												
Objective 3	Develop protocol and obtain ethical approval												
	Focus Groups and qualitative data analysis												
	Communication interface development												
	Write up and submit publication												
	Write up and submit PhD thesis												
Training													
Epidemiology and Medical Statistics Course													
Introduction to Qualitative Research and PPI Training													
Statistical Computing Courses													
Risk Prediction Courses													
Genetic Epidemiology Course													
Kings Fund Leadership Course													
Advanced Qualitative Research Course													
Prognosis Research in Healthcare Course													
ESDR Future Leaders Academy													
Genomic Counselling Skills Course													
Digital Health and Implementation Science Courses													

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